

SIMATS ENGINEERING

ASSESSMENT TOOL 2 – PORT SCANNING CHALLENGE

CO3: Analyze network services using port scanning, ping sweeps, and enumeration techniques across different operating systems to identify vulnerabilities.

COURSE NAME: Ethical Hacking

COURSE CODE: ITA1408

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Register No: 192411119_____

Date: 11.08.2026_____

Lab Environment: Kali Linux + instructor-provided authorized targets.

QUESTION 1: 1. NMAP – SERVER PORT SCANNING: 192.168.1.20

QUESTION: Using Nmap, perform a security assessment on the server 192.168.1.20. Identify all open ports and running services. Analyze the purpose of each discovered service and recommend security measures to reduce the attack surface of the system.

AIM

Using Nmap, perform a security assessment on the server 192.168.1.20. Identify all open ports and running services. Analyze the purpose of each discovered service and recommend security measures to reduce the attack surface of the system.

REQUIREMENTS

Kali Linux, the specified target, the appropriate enumeration tool, terminal access, and permission to test the target.

PROCEDURE / COMMANDS

```
# Step 1 - Check connectivity
ping -c 4 192.168.1.20

# Step 2 - Basic TCP scan
nmap 192.168.1.20

# Step 3 - Scan all TCP ports
sudo nmap -sS -p- 192.168.1.20 -oN Q1_allports.txt

# Step 4 - Detect service and version
sudo nmap -sV 192.168.1.20 -oN Q1_services.txt

# Step 5 - OS fingerprinting
sudo nmap -O 192.168.1.20 -oN Q1_os.txt

# Step 6 - Combined authorized assessment
sudo nmap -sS -sV -O 192.168.1.20 -oN Q1_final.txt
```

EXECUTION EVIDENCE

Paste 2–3 actual screenshots here: (1) command/target, (2) main output, and (3) saved result if applicable.

DETAILED OBSERVATION / ANALYSIS

The purpose of this assessment is to identify the server's exposed attack surface. Nmap reports open ports and, with service detection, provides information about the applications listening on those ports. For example, a web service may provide application access, SSH may provide remote administration, and a database service may provide data access. The exact services must be taken from the student's actual Nmap output. Each open service should have a valid business or laboratory purpose. An unnecessary service increases attack surface because it creates another network entry point. Services should therefore be minimized and restricted to required networks and users. The assessment should also consider software versions. An old version does not automatically prove a vulnerability, but it indicates the need for patch and configuration review. Management services should not normally be exposed broadly, and database services should be restricted to application servers or trusted management hosts.

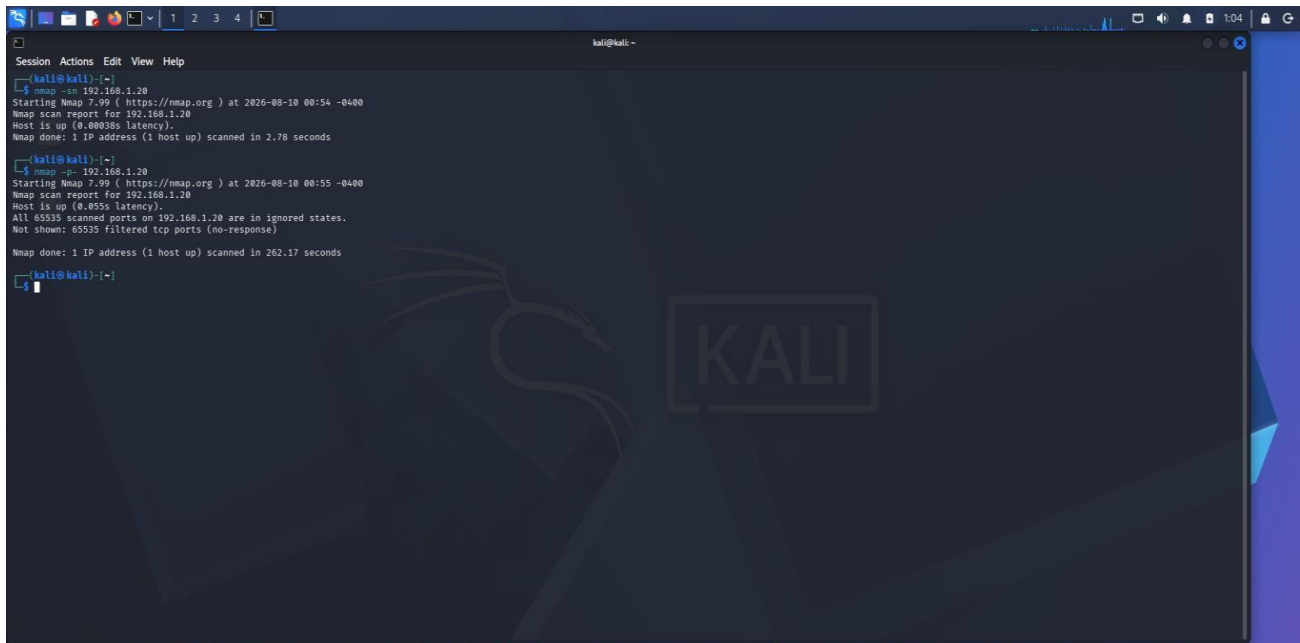
RISK ANALYSIS

Create a table with: Port, Protocol, Service, Version, Purpose, Exposure, Risk and Recommendation. High risk: an unnecessary or weakly protected remotely accessible service with significant impact. Medium risk: an exposed service that is required but should be restricted or hardened. Low risk: a required, appropriately restricted service with limited information exposure. Recommended controls: 1. Disable unused services. 2. Apply security updates and patches. 3. Restrict ports using host/network firewalls. 4. Allow management services only from authorized hosts. 5. Replace insecure protocols with encrypted alternatives. 6. Monitor logs and network connections. 7. Re-scan after hardening to verify that unnecessary ports are closed.

RECOMMENDATIONS

Create a table with: Port, Protocol, Service, Version, Purpose, Exposure, Risk and Recommendation. High risk: an unnecessary or weakly protected remotely accessible service with significant impact. Medium risk: an exposed service that is required but should be restricted or hardened. Low risk: a required, appropriately restricted service with limited information exposure. Recommended controls: 1. Disable unused services. 2. Apply security updates and patches. 3.

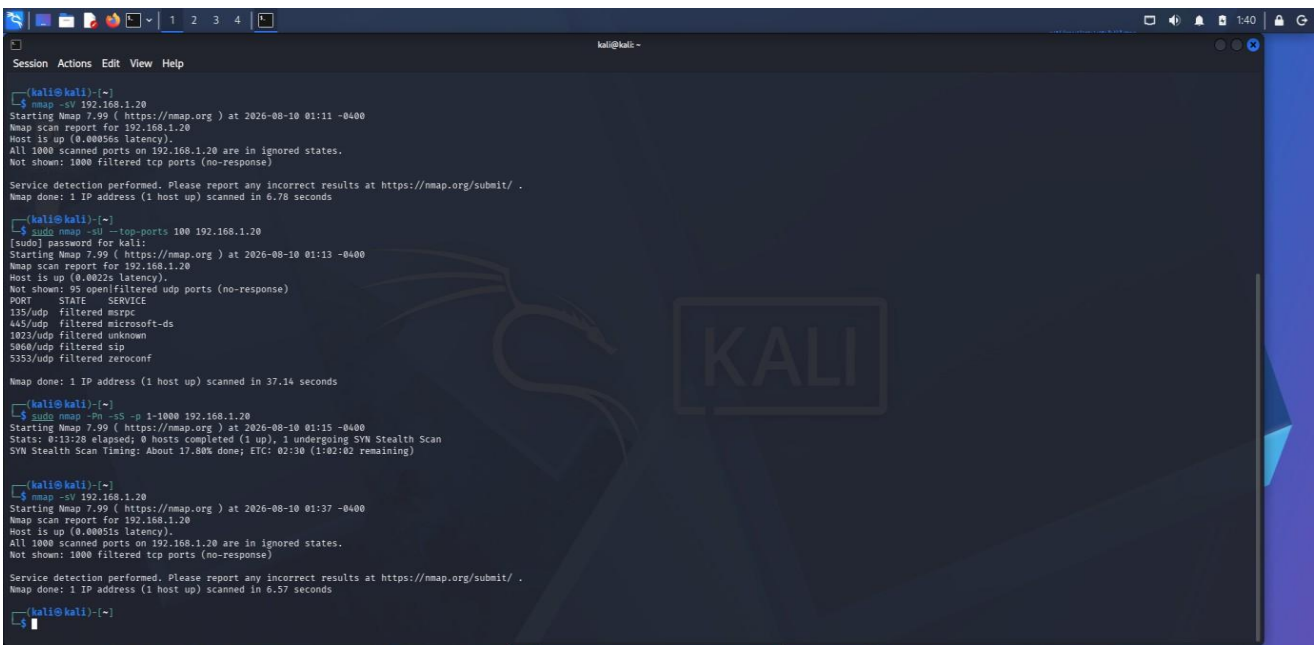
Restrict ports using host/network firewalls. 4. Allow management services only from authorized hosts. 5. Replace insecure protocols with encrypted alternatives. 6. Monitor logs and network connections. 7. Re-scan after hardening to verify that unnecessary ports are closed.



```
kali@kali:~$ nmap -p- 192.168.1.20
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-10 00:54 -0400
Nmap scan report for 192.168.1.20
Host is up (0.00036s latency).
Nmap done: 1 IP address (1 host up) scanned in 2.78 seconds

kali@kali:~$ nmap -p- 192.168.1.20
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-10 00:55 -0400
Nmap scan report for 192.168.1.20
Host is up (0.055s latency).
All 45535 scanned ports on 192.168.1.20 are in ignored states.
Not shown: 65535 filtered tcp ports (no-response)
Nmap done: 1 IP address (1 host up) scanned in 262.17 seconds

kali@kali:~$
```



```
kali@kali:~$ nmap -p- 192.168.1.20
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-10 01:11 -0400
Nmap scan report for 192.168.1.20
Host is up (0.00056s latency).
All 1000 scanned ports on 192.168.1.20 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 6.78 seconds

kali@kali:~$ sudo nmap -sU -top-ports 100 192.168.1.20
[sudo] password for kali:
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-10 01:13 -0400
Nmap scan report for 192.168.1.20
Host is up (0.0022s latency).
Not shown: 95 open/filtered udp ports (no-response)
PORT      STATE SERVICE
135/udp    filtered masqp
445/udp    filtered microsoft-ds
1023/udp   filtered unknown
5060/udp   filtered sip
5353/udp   filtered zeroconf
Nmap done: 1 IP address (1 host up) scanned in 37.14 seconds

kali@kali:~$ sudo nmap -Ph -pS -p 1-1000 192.168.1.20
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-10 01:15 -0400
Stats: 0:13:28 elapsed; 0 hosts completed (1 up); 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 17.80% done; ETC: 02:30 (1:02:02 remaining)

kali@kali:~$ nmap -sV 192.168.1.20
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-10 01:37 -0400
Nmap scan report for 192.168.1.20
Host is up (0.00051s latency).
All 1000 scanned ports on 192.168.1.20 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 6.57 seconds

kali@kali:~$
```

RESULT

Result: The server's exposed ports and services were identified using Nmap. The results provide an inventory of the network attack surface. Unnecessary services should be disabled and required services should be patched, restricted and monitored.

QUESTION 2: 2. NIKTO – WEB SERVER ASSESSMENT: 192.168.1.30

QUESTION: Using Nikto, assess the web server hosted on 192.168.1.30. Identify potential web server misconfigurations, outdated components, or exposed files and directories. Discuss the security risks associated with your findings and suggest appropriate mitigation strategies.

AIM

Using Nikto, assess the web server hosted on 192.168.1.30. Identify potential web server misconfigurations, outdated components, or exposed files and directories. Discuss the security risks associated with your findings and suggest appropriate mitigation strategies.

REQUIREMENTS

Kali Linux, the specified target, the appropriate enumeration tool, terminal access, and permission to test the target.

PROCEDURE / COMMANDS

```
# Step 1 - Identify web ports
nmap -sV -p 80,443 192.168.1.30

# Step 2 - Run Nikto against authorized HTTP service
nikto -h http://192.168.1.30 -output Q2_nikto.txt

# If HTTPS is the authorized service
nikto -h https://192.168.1.30 -output Q2_nikto_https.txt

# Optional Nmap service confirmation
sudo nmap -sV -p 80,443 192.168.1.30 -oN Q2_web_services.txt
```

EXECUTION EVIDENCE

Paste 2–3 actual screenshots here: (1) command/target, (2) main output, and (3) saved result if applicable.

DETAILED OBSERVATION / ANALYSIS

Nikto is used to identify common web-server configuration problems and potentially exposed resources. The output may indicate server information, default files, directory listing, insecure headers, outdated software indicators or other configuration issues. Each result must be interpreted carefully because automated scanner findings require validation. An exposed default page or unnecessary file can disclose information about the application and server. Directory listing may reveal files that were not intended to be public. Missing security headers can weaken browser-side protections. An outdated component may contain known weaknesses and should be reviewed against the vendor's current supported release. The assessment should distinguish between confirmed configuration issues and informational observations. The student's report should quote only the relevant result and explain its impact rather than copying the complete scanner output.

RISK ANALYSIS

For every important Nikto finding, record: Finding, Evidence, Security Impact, Severity and Mitigation. Typical mitigation: 1. Remove default/test files and unnecessary directories. 2. Disable directory listing where not required. 3. Update the web server and frameworks. 4. Remove unnecessary version disclosure. 5. Configure appropriate security headers. 6. Restrict administrative interfaces. 7. Use HTTPS for sensitive traffic. 8. Review server permissions so the web process cannot access unrelated files.

RECOMMENDATIONS

For every important Nikto finding, record: Finding, Evidence, Security Impact, Severity and Mitigation. Typical mitigation: 1. Remove default/test files and unnecessary directories. 2. Disable directory listing where not required. 3. Update the web server and frameworks. 4. Remove unnecessary version disclosure. 5. Configure appropriate security headers. 6. Restrict administrative interfaces. 7. Use HTTPS for sensitive traffic. 8. Review server permissions so the web process cannot access unrelated files.

```
No mail.
msfadmin@metasploitable:~$ sudo ifconfig eth0 192.168.56.103 netmask 255.255.255.0 up
[sudo] password for msfadmin:
msfadmin@metasploitable:~$ ifconfig
eth0      Link encap:Ethernet  HWaddr 08:00:27:73:fb:10
          inet addr:192.168.56.103  Bcast:192.168.56.255  Mask:255.255.255.0
          inet6 addr: fe80::a00:27ff:fe73:fb10/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:519 errors:0 dropped:0 overruns:0 frame:0
          TX packets:77 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:34902 (34.0 KB)  TX bytes:10419 (10.1 KB)
          Base address:0xd020 Memory:f0200000-f0220000

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128 Scope:Host
          UP LOOPBACK RUNNING  MTU:16436  Metric:1
          RX packets:118 errors:0 dropped:0 overruns:0 frame:0
          TX packets:118 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:31841 (31.0 KB)  TX bytes:31841 (31.0 KB)

msfadmin@metasploitable:~$
```

RESULT

Result: Nikto was used to enumerate the authorized web server and identify possible configuration weaknesses. The findings should be validated and corrected according to their actual severity and business impact.

QUESTION 3: 3. NETCAT – SERVICE BANNER COLLECTION: 192.168.1.40

QUESTION: Using Netcat (nc), investigate the host 192.168.1.40 by connecting to open ports and collecting available service banners. Analyze the information obtained and explain how attackers could use banner information during the reconnaissance phase of an attack.

AIM

Using Netcat (nc), investigate the host 192.168.1.40 by connecting to open ports and collecting available service banners. Analyze the information obtained and explain how attackers could use banner information during the reconnaissance phase of an attack.

REQUIREMENTS

Kali Linux, the specified target, the appropriate enumeration tool, terminal access, and permission to test the target.

PROCEDURE / COMMANDS

```
# Step 1 - Identify open services
nmap -sV 192.168.1.40

# Step 2 - Connect to a known open TCP port in the lab
nc -nv 192.168.1.40 <PORT>

# Example for an authorized HTTP service
nc -nv 192.168.1.40 80

# After connection, type a simple HTTP request:
GET / HTTP/1.0
Host: 192.168.1.40

# Exit Netcat
# Press Ctrl+C when finished
```

EXECUTION EVIDENCE

Paste 2–3 actual screenshots here: (1) command/target, (2) main output, and (3) saved result if applicable.

DETAILED OBSERVATION / ANALYSIS

A service banner is information returned by a network service when a connection is established. It may identify the service type, product, version, protocol or other implementation details. The exact banner should be copied from the student's actual lab output. From a defensive perspective, banner information helps administrators identify what software is exposed. From an attacker's reconnaissance perspective, detailed version information can help narrow down which software-specific weaknesses to investigate. This is why unnecessary information disclosure should be minimized. Banner collection itself does not prove that a system is vulnerable. The correct conclusion is that the information may increase reconnaissance value and should be considered together with patch level, configuration, authentication and network exposure.

RISK ANALYSIS

Recommended controls: 1. Keep services fully patched. 2. Reduce unnecessary version disclosure. 3. Disable unused services. 4. Restrict access to management services. 5. Use firewalls/ACLs to limit source networks. 6. Monitor unusual connection attempts. 7. Maintain an accurate software inventory. In the report, include the port used, banner observed, what the banner reveals, why it matters, risk level and mitigation.

RECOMMENDATIONS

Recommended controls: 1. Keep services fully patched. 2. Reduce unnecessary version disclosure. 3. Disable unused services. 4. Restrict access to management services. 5. Use firewalls/ACLs to limit source networks. 6. Monitor unusual connection attempts. 7. Maintain an accurate software inventory. In the report, include the port used, banner observed, what the banner reveals, why it matters, risk level and mitigation.

```
Session Actions Edit View Help
zsh: corrupt history file /home/kali/.zsh_history

(kali@kali)~$ ping -A 192.168.56.102
PING 192.168.56.102: 56(84) bytes of data:
64 bytes from 192.168.56.102: icmp_seq=1 ttl=64 time=0.043 ms
64 bytes from 192.168.56.102: icmp_seq=2 ttl=64 time=0.043 ms
64 bytes from 192.168.56.102: icmp_seq=3 ttl=64 time=0.043 ms

192.168.56.102 ping statistics:
 4 packets transmitted, 4 received, 0% packet loss, time 3014ms
rtt min/avg/max/mdev = 7.635/23.093/51.377/17.743 ms

(kali@kali)~$ nc -v 192.168.56.102 1-1000
192.168.56.102: inverse host lookup failed: Unknown host
(UNKNOWN) [192.168.56.102] 1 (ssh) open
(UNKNOWN) [192.168.56.102] 21 (ftp) open
(UNKNOWN) [192.168.56.102] 22 (ssh) open
(UNKNOWN) [192.168.56.102] 23 (telnet) open
(UNKNOWN) [192.168.56.102] 25 (smtp) open
(UNKNOWN) [192.168.56.102] 80 (http) open
(UNKNOWN) [192.168.56.102] 443 (https) open

(kali@kali)~$ nc -v 192.168.56.102 21
192.168.56.102: inverse host lookup failed: Unknown host
(UNKNOWN) [192.168.56.102] 21 (ftp) open
>>> C

(kali@kali)~$ nc -v 192.168.56.102 22
192.168.56.102: inverse host lookup failed: Unknown host
(UNKNOWN) [192.168.56.102] 22 (ssh) open
SSH-2.0-OpenSSH_4.7p1 Debian-Subuntui
>>> C

(kali@kali)~$ nc -v 192.168.56.102 80
192.168.56.102: inverse host lookup failed: Unknown host
GET / HTTP/1.0

HTTP/1.1 200 OK
Date: Tue, 11 Aug 2026 04:59:44 GMT
Server: Apache/2.2.8 (Ubuntu) DAV/2
```

```
Session Actions Edit View Help
Terminal Emulator
Use the command line

SSH-2.0-OpenSSH_4.7p1 Debian-Subuntui
>>> C

(kali@kali)~$ nc -v 192.168.56.102 80
192.168.56.102: inverse host lookup failed: Unknown host
(UNKNOWN) [192.168.56.102] 80 (http) open
GET / HTTP/1.0

HTTP/1.1 200 OK
Date: Tue, 11 Aug 2026 04:59:44 GMT
Server: Apache/2.2.8 (Ubuntu) DAV/2
X-Powered-By: PHP/5.2.4-2ubuntu5.10
Connection: close
Content-Type: text/html

<html><head><title>Metasploitable2 - Linux</title></head><body>
<pre>

  ITA14EXPLOITABLE2

Warning: Never expose this VM to an untrusted network!
Contact: msfdev[at]metasploit.com
Login with msfadmin/msfadmin to get started

</pre>
<ul>
<li><a href="/twiki/">TWiki</a></li>
<li><a href="/phpMyAdmin/">phpMyAdmin</a></li>
<li><a href="/mutillidae/">Mutillidae</a></li>
<li><a href="/dvwa/">DVWA</a></li>
<li><a href="/dav/">WebDAV</a></li>
</ul>
</body>
</html>

(kali@kali)~$ nc -v 192.168.56.102 25
192.168.56.102: inverse host lookup failed: Unknown host
(UNKNOWN) [192.168.56.102] 25 (smtp) open
220 metasploitable.localdomain ESMTP Postfix (Ubuntu)
```

RESULT

Result: Netcat was used to collect service-banner information from an authorized open port. The information was analyzed from both defensive and reconnaissance perspectives, with emphasis on reducing unnecessary information disclosure.

QUESTION 4: 4. TELNET – REMOTE ADMINISTRATION CHECK: 192.168.1.50

QUESTION: Using Telnet, examine the server 192.168.1.50 to determine whether remote administration services are accessible. Identify any security concerns related to the discovered services and recommend more secure alternatives where appropriate.

AIM

Using Telnet, examine the server 192.168.1.50 to determine whether remote administration services are accessible. Identify any security concerns related to the discovered services and recommend more secure alternatives where appropriate.

REQUIREMENTS

Kali Linux, the specified target, the appropriate enumeration tool, terminal access, and permission to test the target.

PROCEDURE / COMMANDS

```
# Step 1 - Check the management ports
nmap -sV -p 22,23 192.168.1.50

# Step 2 - If Telnet is intentionally enabled in the authorized lab
telnet 192.168.1.50 23

# Step 3 - If SSH is available, identify it as the secure alternative
ssh -v <authorized-user>@192.168.1.50

# Do not use unknown or unauthorized credentials.
```

EXECUTION EVIDENCE

Paste 2–3 actual screenshots here: (1) command/target, (2) main output, and (3) saved result if applicable.

DETAILED OBSERVATION / ANALYSIS

Remote administration provides a way to manage a server without physical access, but it must be strongly protected. Telnet is an insecure legacy protocol because it does not provide the encryption expected for remote administration. Credentials and session data can therefore be exposed to interception on an untrusted network. SSH is normally preferred because it provides encrypted communication and supports stronger authentication mechanisms. However, simply having SSH open does not make a system secure: access should be restricted, authentication should be hardened, and logging should be enabled. The report should state whether ports 22 and/or 23 are open based on actual Nmap output, describe the purpose of the service, and explain the security implications.

RISK ANALYSIS

Recommended controls: 1. Disable Telnet when it is not required. 2. Use SSH for authorized remote administration. 3. Restrict SSH access using firewall rules or ACLs. 4. Prefer strong authentication and, where appropriate, key-based authentication. 5. Disable unnecessary remote-management accounts. 6. Keep the SSH server patched. 7. Monitor authentication and administrative logs. 8. Place management interfaces on a restricted management network where possible.

RECOMMENDATIONS

Recommended controls: 1. Disable Telnet when it is not required. 2. Use SSH for authorized remote administration. 3. Restrict SSH access using firewall rules or ACLs. 4. Prefer strong authentication and, where appropriate, key-based authentication. 5. Disable unnecessary remote-management accounts. 6. Keep the SSH server patched. 7. Monitor authentication and administrative logs. 8. Place management interfaces on a restricted management network where possible.


```
Session [1] Terminal Emulator
Use the command line

zsh: corrupt history file /home/kali/.zsh_history
(kali@kali)~$ telnet --help
Usage: telnet [OPTION...] [HOST [PORT]]
Login to remote system HOST (optionally, on service port PORT)

General options:
  -4, --ipv4                use only IPv4
  -6, --ipv6                use only IPv6
  -8, --binary              use an 8-bit data transmission
  -a, --login               attempt automatic login
  -b, --bind=ADDRESS        bind to specific local ADDRESS
  -c, --no-rc               do not read the user's .telnetrc file
  -d, --debug               turn on debugging
  -e, --escape=CHAR         use CHAR as an escape character
  -E, --no-escape           use no escape character
  -k, --no-login            do not automatically login to the remote system
  -l, --user=USER           attempt automatic login as USER
  -L, --binary-output       use an 8-bit data transmission for output only
  -n, --traces=FILE         record trace information into FILE
  -r, --rlogin              use a user-interface similar to rlogin

Encryption control:
  -x, --encrypt             encrypt the data stream, if possible

Authentication and Kerberos options:
  -k, --realm=REALM         obtain tickets for the remote host in REALM
                           instead of the remote host's realm
  -X, --disable-auth=ATYPE  disable type ATYPE authentication
  -7, --help                give this help list
  -u, --usage                give a short usage message
  -V, --version              print program version

Mandatory or optional arguments to long options are also mandatory or optional
for any corresponding short options.

Report bugs to <bug-inetutils@gnu.org>.

(kali@kali)~$ telnet 192.168.56.102 23
Trying 192.168.56.102...
Connected to 192.168.56.102.
Escape character is '^['.
```

```
kali@kali -
Firefox ESR
Browse the World Wide Web

zsh: corrupt history file /home/kali/.zsh_history
(kali@kali)~$ telnet 192.168.56.102 22
Trying 192.168.56.102...
Connected to 192.168.56.102.
Escape character is '^['.
SSH-2.0-OpenSSH_4.7p1 Debian-8ubuntu1
*)
telnet> quit
Connection closed.

(kali@kali)~$
```

RESULT

Result: The remote administration exposure of 192.168.1.50 was examined. Any Telnet exposure should be removed where possible, and remote management should be restricted and performed using secure, encrypted administration mechanisms.

QUESTION 5: 5. NETDISCOVER – HOST DISCOVERY: 192.168.1.1–50

QUESTION: Using Netdiscover, perform host discovery on the network segment 192.168.1.1–50. Identify active hosts on the network and analyze how the discovered information could assist an attacker in planning further enumeration or exploitation activities.

AIM

Using Netdiscover, perform host discovery on the network segment 192.168.1.1–50. Identify active hosts on the network and analyze how the discovered information could assist an attacker in planning further enumeration or exploitation activities.

REQUIREMENTS

Kali Linux, the specified target, the appropriate enumeration tool, terminal access, and permission to test the target.

PROCEDURE / COMMANDS

```
# Step 1 - Identify your network interface
ip addr

# Step 2 - Discover hosts on the authorized lab segment
sudo netdiscover -i <INTERFACE> -r 192.168.1.0/24

# Review the output and record only the authorized range
# 192.168.1.1 through 192.168.1.50
```

EXECUTION EVIDENCE

Paste 2–3 actual screenshots here: (1) command/target, (2) main output, and (3) saved result if applicable.

DETAILED OBSERVATION / ANALYSIS

Host discovery is the reconnaissance stage in which active systems are identified before more detailed enumeration. Netdiscover can display information such as IP addresses, MAC addresses and vendor information on a local Ethernet network. The discovered hosts can help create a basic network inventory. An attacker could use the information to decide which systems deserve further investigation, for example servers, workstations, gateways or other infrastructure. In an authorized assessment, this same information helps defenders identify unexpected devices and improve asset inventory. Only the authorized lab segment should be included in the report. The output should be summarized rather than treated as proof that a host is vulnerable.

RISK ANALYSIS

Recommended controls: 1. Segment sensitive systems using VLANs or other network segmentation. 2. Enable host firewalls. 3. Restrict unnecessary network services. 4. Monitor for unexpected discovery/scanning activity. 5. Maintain an accurate asset inventory. 6. Use network access controls where appropriate. 7. Separate management networks from normal user networks. Create a table with IP Address, MAC Address, Vendor, Host Type/Role if known, and Security Observation.

RECOMMENDATIONS

Recommended controls: 1. Segment sensitive systems using VLANs or other network segmentation. 2. Enable host firewalls. 3. Restrict unnecessary network services. 4. Monitor for unexpected discovery/scanning activity. 5. Maintain an accurate asset inventory. 6. Use network access controls where appropriate. 7. Separate management networks from normal user networks. Create a table with IP Address, MAC Address, Vendor, Host Type/Role if known, and Security Observation.

```
Firefox ESR
Browse the World Wide Web

zsh - sshd@kali:~$ cat /home/kali/.ssh_history
[Kali@kali]~$
1: ip addr
link/loopback 08:00:27:00:00:00 brd 00:00:00:00:00:00
valid_lft 2/0 scope host lg
inet 127.0.0.1 scope host preferred_lft forever
link/ether 08:00:27:16:3b:05 brd ff:ff:ff:ff:ff:ff
2: eth0
BROADCAST,MULTICAST,UP,LOWER_UP mtu 1500 qdisc fq_codel state UP group default qlen 1000
link/ether 08:00:27:16:3b:05 brd ff:ff:ff:ff:ff:ff
valid_lft 835955sec preferred_lft 835955sec
inet 192.168.56.122 brd 192.168.56.255 scope global dynamic noprefixroute eth0
valid_lft 847915sec preferred_lft 127915sec
inet 192.168.56.122 brd 192.168.56.255 scope link noprefixroute
3: eth1
link/ether 08:00:27:16:3b:05 brd ff:ff:ff:ff:ff:ff
valid_lft 4055sec preferred_lft 4055sec
inet 192.168.56.123 brd 192.168.56.255 scope global dynamic noprefixroute eth1
link/ether 08:00:27:16:3b:05 brd ff:ff:ff:ff:ff:ff
4: docker0
NO-CARRIER,BROADCAST,MULTICAST,UP mtu 1500 qdisc noqueue state DOWN group default
link/ether 00:00:00:00:00:00 brd 00:00:00:00:00:00
valid_lft forever preferred_lft forever
[Kali@kali]~$
$ sudo netdiscover -i eth1 -r 192.168.56.0/24
```

4 Captured ARP Req/Rep packets, from 3 hosts. Total size: 240									
IP	AT	MAC Address	Count	Len	MAC Vendor	Hostname			
192.168.56.100	08:00:00:00:00:00	08:00:27:16:3b:c3	1	120	PCS Systemtechnik GmbH				
192.168.56.102	08:00:00:00:00:00	08:00:27:16:3b:0a	1	60	Unknown Vendor				
192.168.56.102	08:00:27:16:3b:0a	08:00:27:16:3b:10	1	60	PCS Systemtechnik GmbH				

RESULT

Result: Netdiscover identified active systems within the authorized lab segment. The discovered information can be used for legitimate asset inventory and can also increase reconnaissance value to an attacker, so network segmentation, monitoring and access controls are recommended.

CONCLUSION

The practical assessment demonstrates the importance of reconnaissance and service enumeration in understanding network attack surface. Nmap identifies exposed ports and services, Nikto reviews web-server exposure, Netcat demonstrates service-banner collection, Telnet highlights remote-administration risks, and Netdiscover identifies active hosts. The main defensive principle is to minimize exposure: disable unnecessary services, patch software, restrict management access, use encrypted protocols, segment networks, reduce information leakage and monitor network activity.

Code of Conduct:

I P.Ranjithkumar(192411119)_ certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: PRKY

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Identification of Port Scanning Factors	25%	Clearly identifies comprehensive and relevant strengths, weaknesses, opportunities, and threats	Identifies most factors with minor gaps	Limited or partially relevant identification	Incorrect or incomplete Port Scanning factors
Analysis	25%	Provides deep, insightful analysis of each Port Scanning component with strong reasoning	Provides good analysis with some depth	Basic analysis with limited reasoning	No meaningful analysis provided
Linkage	20%	Effectively links Port Scanning elements to derive strong strategic insights and implications	Shows some linkage with moderate insights	Weak linkage; limited strategic thinking	No linkage or strategic insight
Application	20%	Applies Port Scanning analysis effectively to the given case or scenario with clear relevance	Applies analysis with minor gaps	Limited application; lacks clarity	No clear application to context
Clarity	10%	Well-structured, clear, and logically presented with proper formatting	Generally clear with minor issues	Some structure but lacks clarity	Poorly organized and difficult to understand

